Stacks

Chapter 6

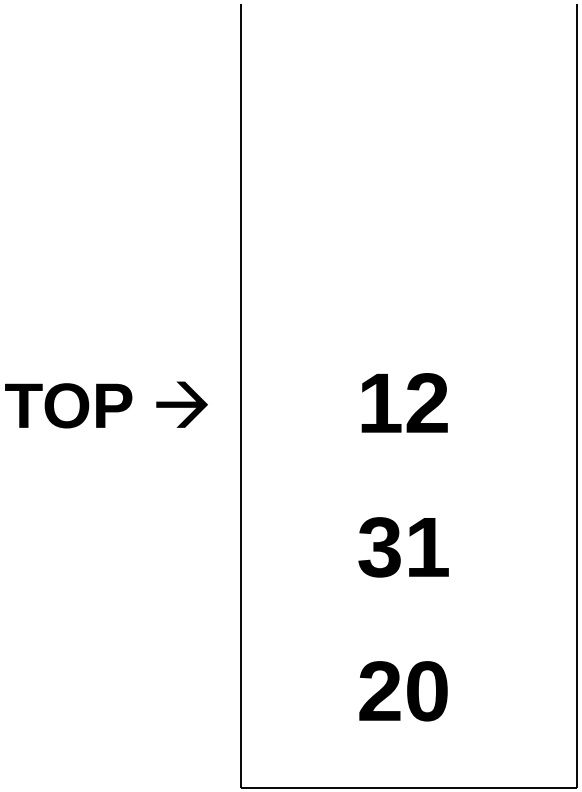
Fundamentals

- A <u>stack</u> is a sequence of data elements (of the same type) arranged one after another conceptually.
- An element can be added to the top of the stack only. (“PUSH”)
- An element can be removed from the top of the stack only. (“POP”)

Applications of Stacks

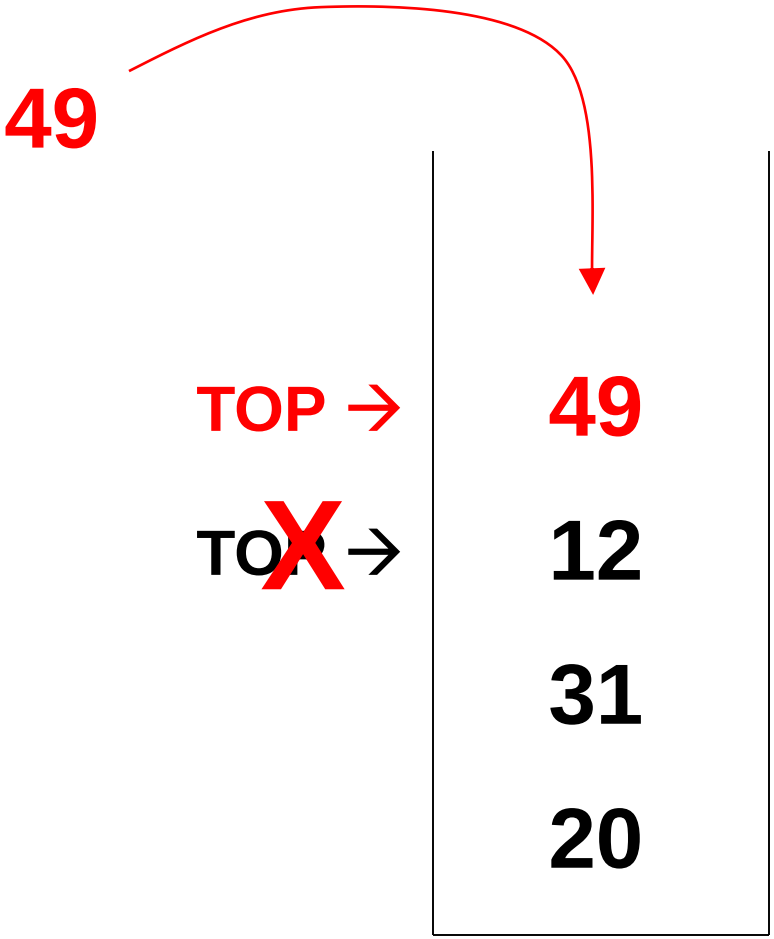
- Operating systems
  - keeping track of method calls in a running program
- Compilers
  - conversion of arithmetic expressions to machine code

Conceptual Picture

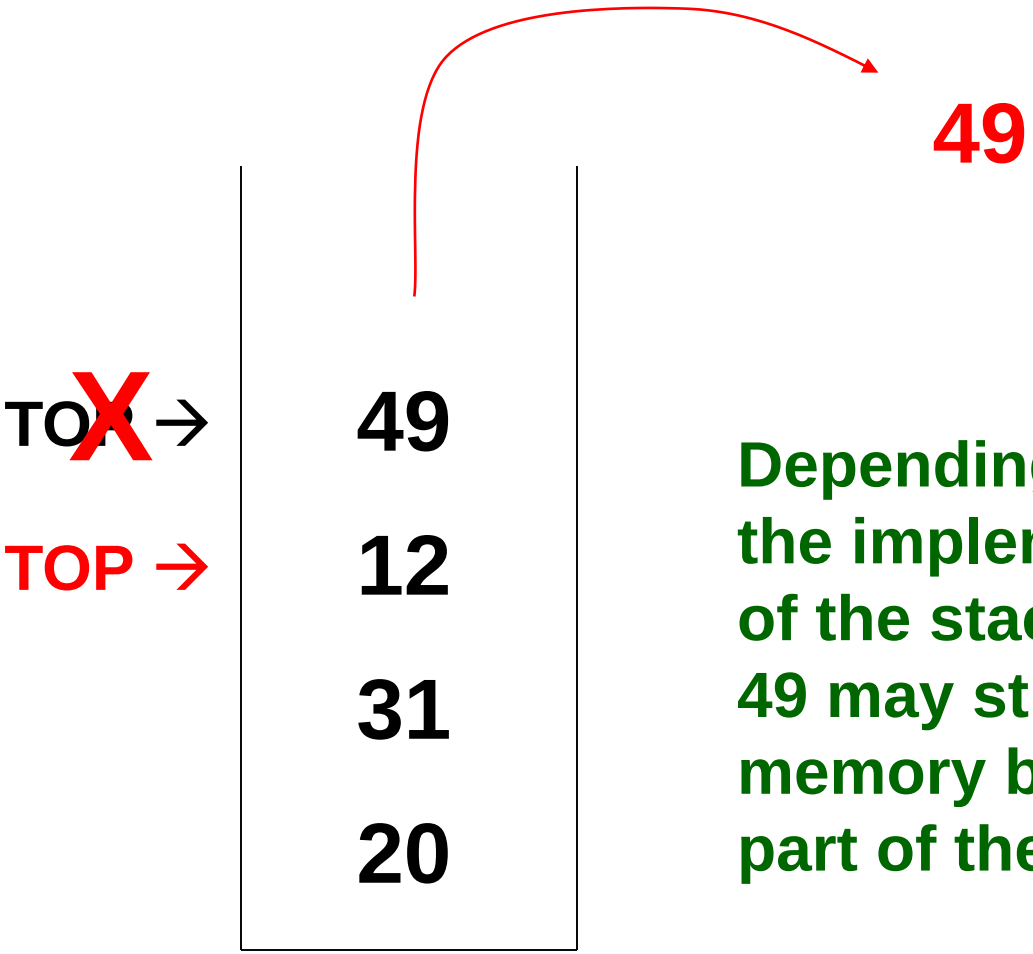


Depending on the implementation of a stack, it may or may not have a maximum capacity.

PUSH



POP



Depending on the implementation of the stack, the 49 may still be in memory but is not part of the stack.

Implementation of a Stack

- **ARRAYS**

12	**31**	**20**				

top

20	**31**	**12**				

top

Which is more efficient?

Basic Stack Operations

- **Constructor – create an empty stack**
- **isEmpty – is the stack empty?**
- **push – push an element on to the top of the stack (if the stack is not full)**
- **pop – remove the top element from the stack (if the stack is not empty)**

Extended Stack Operations

- **peek – examine the top element of the stack without removing it (if the stack is not empty)**

```
if not isEmpty()
    temp ← pop()
    push(temp)
    return(temp)
```

- **size – return the number of elements on the stack**

How would you implement this using the basic operations?

An IntStack using Arrays

```
public class IntStack implements
  Cloneable {

  public final int CAPACITY = 100;
  private int[] data;
  private int top;

  // IntStack methods (clone not shown)
}
```

IntStack (arrays) (cont'd)

```
public IntStack()
{
    top = -1;
    data = new int[CAPACITY];
}
public boolean isEmpty()
{
    return (top == -1);
}
```

IntStack (arrays) (cont'd)

```
public void push(int item)
{
    if (top == CAPACITY-1)
        throw new FullStackException();
    top++;
    data[top] = item;
}
```

IntStack (arrays) (cont'd)

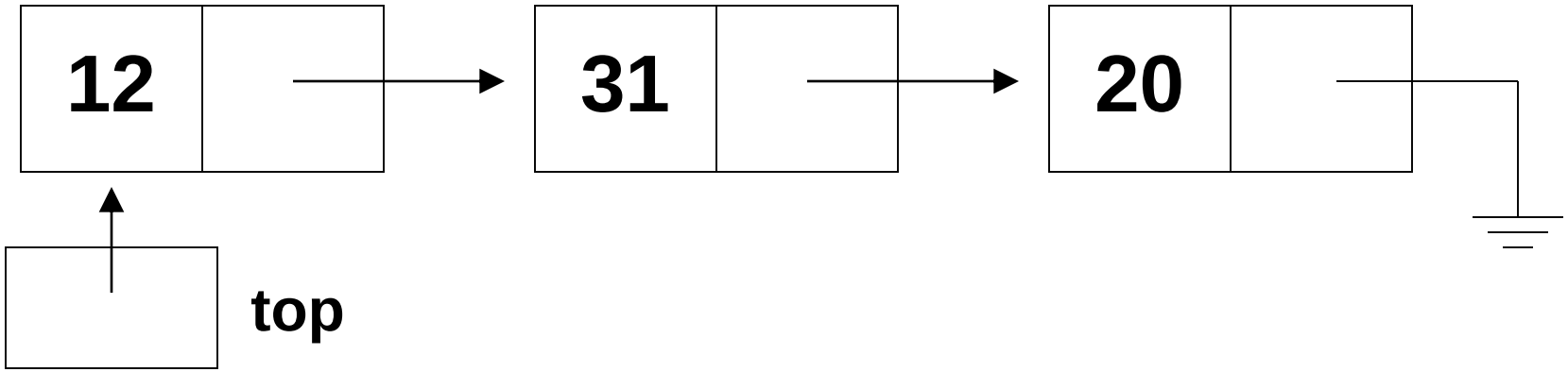
```
public int pop()
{
    int answer;
    if (top == -1) // isEmpty()
        throw new EmptyStackException();
    answer = data[top];
    top--;
    return answer;
}
```

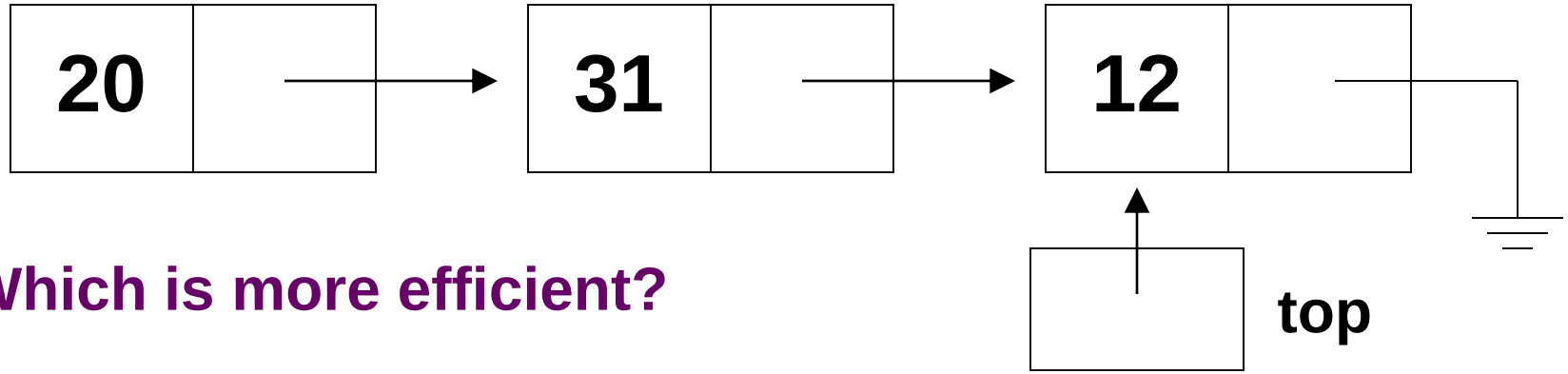
IntStack (arrays) (cont'd)

```
public int peek()
{
  int answer;
  if (top == -1) // isEmpty()
    throw new EmptyStackException();
  answer = data[top];
  return answer;
}
```

Implementation of a Stack

- **LINKED LISTS**





Which is more efficient?

An IntStack using Lists

```
public class IntStack implements
  Cloneable {

  private IntNode top;

  // IntStack methods (clone not shown)
}
```

IntStack (lists) (cont'd)

```
public IntStack()
{
    top = null;
}
public boolean isEmpty()
{
    return (top == null);
}
```

IntStack (lists) (cont'd)

```
public void push(int item)
{
    IntNode newNode
        = new IntNode(item);
    newNode.setLink(top);
    top = newNode;
}
```

Stack Overflow?

IntStack (lists) (cont'd)

```
public int pop()
{
    int answer;
    if (top == null) // isEmpty()
        throw new EmptyStackException();
    answer = top.getData();
    top = top.getLink();
    return answer;
}
```

IntStack (lists) (cont'd)

```
public int peek()
{
  int answer;
  if (top == null) // isEmpty()
    throw new EmptyStackException();
  answer = top.getData();
  return answer;
}
```

Balanced Parentheses

- **An arithmetic expression has balanced parenthesis if and only if:**
  - **the number of left parentheses of each type is equal to the number of right parentheses of each type**
  - **each right parenthesis of a given type matches to a left parenthesis of the same type to its left and all parentheses in between are balanced correctly.**

Examples

- **({A + B} – C)**
  Balanced
- **({A + B) – C}**
  Not balanced
- **({A + B} – [C / D])**
  Balanced
- **(({A + B} – C) / D))**
  Not balanced

Algorithm

CHECK FOR BALANCED PARENTHESES

- **Scan the expression from left to right.**

  For each left parenthesis that is found, push on the stack.

  For each right parenthesis that is found,

  If the stack is empty, return false (too many right parentheses)

  Otherwise, pop the top parenthesis from the stack:

  If the left and right parentheses are of the same type, discard. Otherwise, return false.

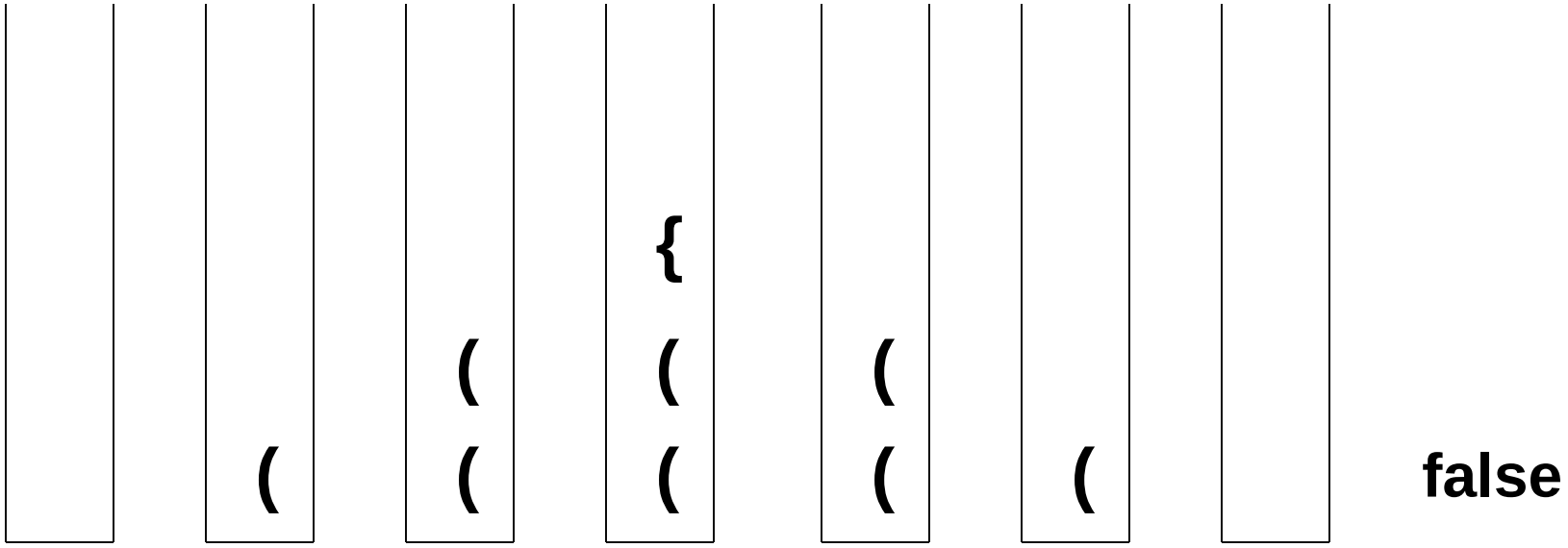
Algorithm (cont'd)

CHECK FOR BALANCED PARENTHESES

- **If the stack is empty when the scan is complete, return true.**
  Otherwise, return false. (too many left parentheses)

Trace

- **(({A + B} – C) / D))**
- **Stack trace:**



Evaluating Expressions

- **An expression is fully parenthesized if every operator has a pair of balanced parentheses marking its left and right operands.**
- **Not fully-parenthesized:**
  3 * (5 + 7) – 9 **(2 - 4) * (5 - 7) + 8**
- **Fully-parenthesized:**
  ((3 * (5 + 7)) – 9) **(((2 - 4)*(5 - 7)) + 8)**

General Idea

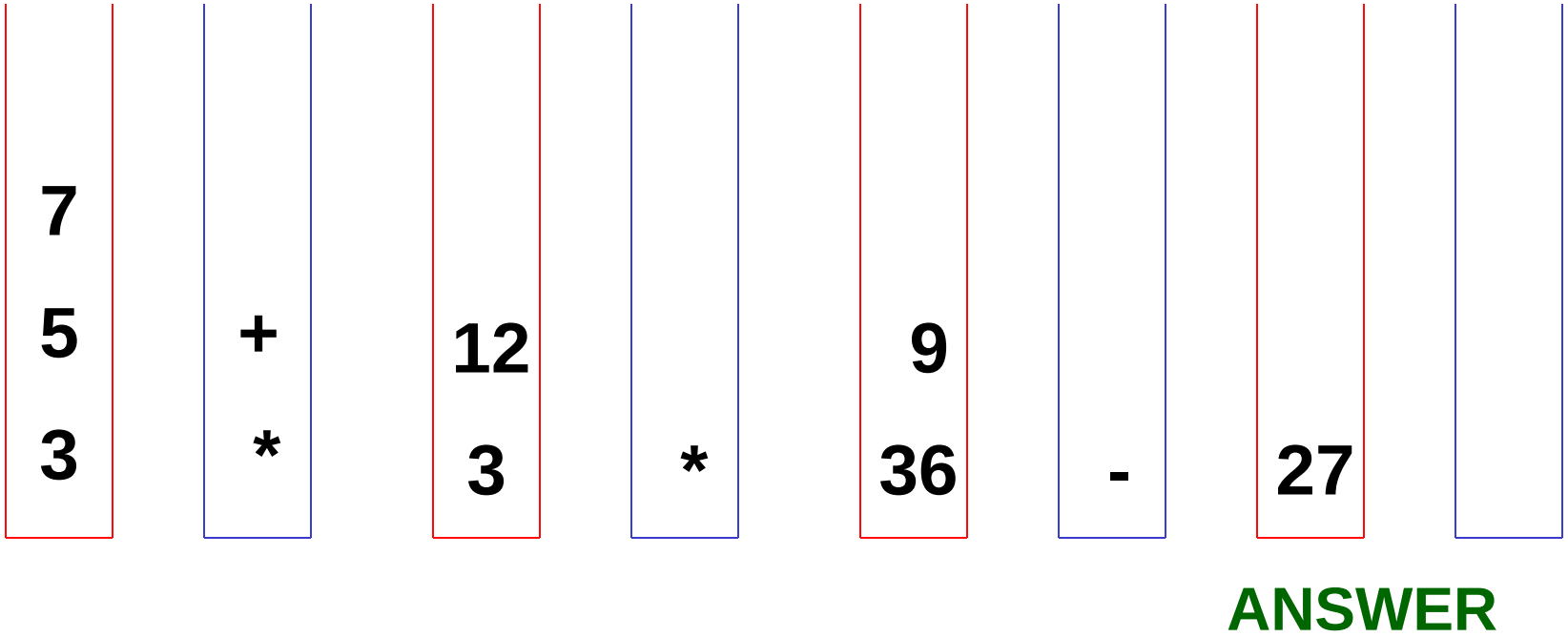
- **The first operation to perform is surrounded by the innermost set of balanced parentheses.**
- **Example: ((3 * (5 + 7)) – 9) First op: +**
- **By reading expression from left to right, first operator comes immediately before first right parenthesis.**
- **Replace that subexpression with its result and search for next right parenthesis, etc.**
- **Example: ((3 * 12) – 9) = (36 – 9) = 27**

General Idea (cont'd)

- **How do we keep track of operands and operators as we read past them in the expression from left to right?**
- **Use two stacks:**
  one for operands and one for operators.
- **When we encounter a right parenthesis, pop off one operator and two operands, perform the operation, and push the result back on the operand stack.**

Trace

- **((3 * (5 + 7)) – 9)**
- **Stack traces:** **operands** **operators**



Algorithm

- **Let each operand or operator or parenthesis symbol be a <u>token</u>.**
- **Let NumStack store the operands.**
- **Let OpStack store the operations.**
- **For each token in the input expression do**

  If token = operand, NumStack.push(token)

  If token = operator, OpStack.push(token)

Algorithm (cont'd)

If token = ")",

operand2 ← NumStack.pop()

operand1 ← NumStack.pop()

operator ← OpStack.pop()

result ← operand1 operator operand2

NumStack.push(result)

If token = "(", ignore token

- **After expression is parsed, answer ← NumStack.pop()**

Arithmetic Expressions

- **Infix notation:**
  operator is between its two operands

  3 + 5 (5 + 7) * 9 5 + (7 * 9)
- **Prefix notation:**
  operator precedes its two operands

  + 3 5 * + 5 7 9 + 5 * 7 9
- **Postfix notation:**
  operator follows its two operands

  3 5 + 5 7 + 9 * 5 7 9 * +

NO PAREN-THESES

Precedence of Operators

- **Multiplication and division (higher precedence) are performed before addition and subtraction (lower precedence)**
- **Operators in balanced parentheses are performed before operators outside of the balanced parentheses.**
- **If two operators are of the same precedence, they are evaluated left to right.**

Example

- **Infix expression:**

A + B * (C * D – E / F) / G – H

6 4 1 3 2 5 7

- **What is its prefix equivalent?**

– + A / * B – * C D / E F G H

- **What is its postfix equivalent?**

A B C D * E F / – * G / + H –

1 2 3 4 5 6 7

Evaluating a Postfix Expression

- **Let each operand or operator be a <u>token</u>.**
- **Let NumStack store the operands.**
- **For each token in the input expression do**

  If token = operand, push(token)

  If token = operator,

  operand2 ← pop()

  operand1 ← pop()

  result ← operand1 operator operand2

  push(result)
- **answer ← pop()**

Trace

- **INFIX:** **3 * (5 + 7) – 9**
- **POSTFIX:** **3 5 7 + * 9 -**
- **Stack traces:** **operands**

7			
5	12	9	
3	3	36	27

ANSWER

Translating Infix to Postfix

Fully-Parenthesized Expressions

- **Let each operand, operator, or parenthesis be a token.**
- **Let OpStack store the operators.**
- **Let postfix string P = “” (empty string)**
- **For each token in the input expression do**

  If token = operand, append operand to P

  If token = operator, push(token)

  If token = “)”, append pop() to P

  If token = “(“, ignore

Trace

- **Infix: ((3 * (5 + 7)) – 9)**

Stack (sideways)	Postfix String
empty	3
*	3
*	3 5
* +	3 5
* +	3 5 7
*	3 5 7 +
empty	3 5 7 + *
–	3 5 7 + *
–	3 5 7 + * 9
empty	3 5 7 + * 9 –

Another Example

Infix: **((3 * (5 + 7)) – 9)**

Postfix: **3 5 7 + * 9 –**

Infix: **(((2 – 4)*(5 – 7)) + 8)**

Postfix: **2 4 – 5 7 – * 8 +**

Another Example

Infix: **((3 * (5 + 7)) – 9)**

2 1 3

Postfix: **3 5 7 + * 9 –**

1 2 3

Infix: **(((2 – 4)*(5 – 7)) + 8)**

1 3 2 4

Postfix: **2 4 – 5 7 – * 8 +**

1 2 3 4

Translating Infix to Postfix

General Expressions

- **Define a precedence function**
- ***prec*: token → {0,1,2,3}**
- **let "$" represent empty stack**
- **token precedence**

"$"	**0**
"("	**1**
"+", "-"	**2**
"*", "/"	**3**

Translating Infix to Postfix

General Expressions

- **Let each operand, operator, or parenthesis be a token.**
- **Let OpStack be a character stack that stores the operators or other special symbols ("(" and "$").**
- **Let postfix string P = "" (empty string)**

Translating Infix to Postfix (cont'd)

General Expressions

1. **push("$")**
2. **For each token in the input expression do**

   a. If token = operand,

   append token to P

   b. if token = "(",

   push(token)

Translating Infix to Postfix (cont'd)

General Expressions

c. if token = ")",

topOp ← pop()

while topOp ≠ "("

append topOp to P

topOp ← pop()

Translating Infix to Postfix (cont'd)

General Expressions

d. if token = operator,

topOp ← peek()

while *prec*(topOp) ≥ *prec*(token)

append pop() to P

topOp ← peek()

push(token)

Translating Infix to Postfix (cont'd)

General Expressions

3. At end of infix expression,

topOp ← pop()

while topOp ≠ "$" do

append topOp to P

topOp ← pop()

Trace

A + B * (C * D - E / F) / G – H

- **Stack** ***(sideways)*** **Postfix String**

$	
$	**A**
$ +	**A**
$ +	**A B**
$ + *	**A B**
$ + * (	**A B**
$ + * (	**A B C**

Trace (cont'd)

A + B * (C * D - E / F) / G – H

- **Stack** ***(sideways)*** **Postfix String**

$ + * (	**A B C**
$ + * (*	**A B C**
$ + * (*	**A B C D**
$ + * (-	**A B C D ***
$ + * (-	**A B C D * E**
$ + * (- /	**A B C D * E**
$ + * (- /	**A B C D * E F**

Trace (cont'd)

A + B * (C * D - E / F) / G – H

- **Stack** ***(sideways)***

Stack	Postfix String
$ + * (- /	A B C D * E F
$ + *	A B C D * E F / -
$ + /	A B C D * E F / - *
$ + /	A B C D * E F / - * G
$ -	A B C D * E F / - * G / +
$ -	A B C D * E F / - * G / + H
empty	A B C D * E F / - * G / + H -

A special case: A+B+C*D

A++ and ++A